Understanding Review Helpfulness: Statistical Inference and Text-Based Prediction in Amazon Fine Food Reviews

Group name	*The Data Forge*
Dataset	Stanford SNAP Amazon Fine Food Reviews: `https://snap.stanford.edu/data/web-FineFoods.html`
Source Code	*https://github.com/skabdulkhadeer343/Amazon-Fine-Food-Reviews-Statistical-Analysis*

S.No	Name	Roll no.	Institute email	Contribution
1	*Agithe Yeshwanth*	*MT25059*	*agithe25059@iiitd.ac.in*	Ensure that parsing of raw data, checking for malformed data, detecting duplicate records, cleaning the data, decision-making on documentation, and ensuring that the divide between exploration and confirmation is not based on product IDs.
2	*Shaik Abdul Khadeer*	*MT25086*	*abdul25086@iiitd.ac.in*	Construct and analyze all EDA graphs, summarize rating and helpfulness trends, study the lengths of the reviews and their summaries, and record the research questions and motivations for future analyses.
3	*Vysyaraju Shanmukha Srinivasa Raju*	*MT25090*	*vysyaraju25090@iiitd.ac.in*	Determine whether there is any correlation between rating extremity and helpfulness as well as the nature of the relationship between the length of the review and helpfulness (linear or nonlinear). Conduct sensitivity analyses and state assumptions and limitations.
4	*Balaiah Tarun*	*2022132*	*balaiah22132@iiitd.ac.in*	Check for further explanation from longer summaries, test if the text features can predict the usefulness, carry out the Holm correction as expected, check for over-dispersion and inference assumption, check the dimension of the text features used, and add the project guidelines, findings, and limitations to the README file.

1. Dataset

Dataset Source: Amazon Fine Foods Reviews from Stanford SNAP dataset in the form of text. Each line represents one review. The raw data set file is named `finefoods.txt`.

The reason why this dataset is considered raw: Data entries are maintained as label-text pairs instead of tabular entries. Dataset attributes include product ID, user ID, helpfulness votes, rating score, date of review, review title, and review text. The text can have multiple lines while vote counts are required to be positive numbers.

	Parsed data	After preparation
Instances (rows)	568,454	568,173
Attributes (columns)	14	16
Instances $\times$ attributes	7,958,356	9,090,768

Check for size: $568{,}173 \times 16 = 9{,}090{,}768 \geq 3{,}000{,}000$,. Moreover, the number of exploration reviews for the TF-IDF representation is 454,951 with 20,005 features total, which gives us $454{,}951 \times 20{,}005 = 9{,}101{,}294{,}755$. It is the high-dimensional encoding representation that will be used to show the requirement on the feature scale.

Key features

Feature	Type	Real-world meaning
`Score`	discrete/ordinal	Product rating on scale 1 to 5 stars; observed rating of review.
`ProductId`	nominal	Product identification number; used for product clustering splitting and inference.
`UserId`	nominal	User identification number; not treated as numeric.
`Helpfulness Numerator`	discrete count	Number of helpful votes.
`Helpfulness Denominator`	discrete count	Total number of helpfulness votes on source.
`Helpfulness Rate`	continuous, bounded	Ratio of helpful votes only if both values are valid and the denominator is not zero.
`ReviewLength Chars`	discrete count	Number of Characters in the body of the review after removing spaces.
`SummaryLength Chars`	discrete count	Characters in the review summary after removing spaces..
`ReviewDateUTC`	date/time	Parsed review timestamp in UTC.
`TF−IDF features`	continuous, sparse	Text-derived unigram and bigram weights from summaries and review bodies.

2. Data preparation

Issue found	What we did	Why, and not the alternative
Raw text blocks and continuation lines	Parses labelled fields; splits each field at its first : ensures continuation lines in the review text.	This keeps the review text whole without splitting on punctuation in the text.
Missing profile names and summaries	Preserved missing profile names; missing summaries represented as empty strings in text form.	Absence of metadata does not imply that a review should be thrown away; missing summary is not made-up data.
281 exact duplicate rows	Identified exact duplicates after trimming down selected text variables.	Exact duplicates may result in double-counting of the same observation; records with text similarities among products were not automatically considered duplicates.
Invalid helpfulness counts	Identified the two records that had the number of helpful votes greater than total votes; kept them in the clean data set, but not in the analysis for valid rates.	Ensures that the original data values are not silently fixed while stopping improbable ratios from being used in rate-based tests.
Zero-vote reviews	They kept it, with undefined help rate.	Zero denominator is not a zero help rate because doing that would influence the response negatively.
Long reviews and large vote counts	IQR thresholds were used for diagnostic purposes, but these observations were neither deleted nor capped.	High review lengths and votes can very well be genuine. Capping or deleting would alter the data and the estimand without proof of any errors.
Text representation	TF–IDF built on exploration data alone, limited to 20,000 unigram/bigram features maximum and $\min_d f = 5$.	Building the vocabulary on exploration data prevents using confirmation data for building the feature space.
Product overlap	Separate by product using seed 42.	It prevents duplication of the same product in both the exploratory and confirmatory samples.

3. What changed after preparation

Measure	Before	After
Rows	568,454	568,173
Attributes	14	16
Exact duplicate rows removed	–	281
Missing `ProfileName`	26	26 retained
Missing `Summary`	27	0 represented as empty strings
Zero helpfulness denominator	270,052	269,850
Numerator greater than denominator	2	2 flagged and retained
Valid helpfulness-rate records	–	298,321

In preparation, 281 exact duplicates were eliminated, decreasing the number of rows by about 0.049%. The data set retains the two problematic instances of helpfulness scores while marking them, and does not convert reviews with zero votes to have zero helpfulness scores. The exploration set has 454,951 rows, while the validrate set for exploratory analysis has 239,593 rows. In the case of text modeling, the feature matrix consists of 20,005 features in 454,951 exploration rows, or 9,101,294,755 positions altogether. It is a sparse matrix, so the arithmetic includes both zeros and non-zeros.

4. Basic questions and problem statement

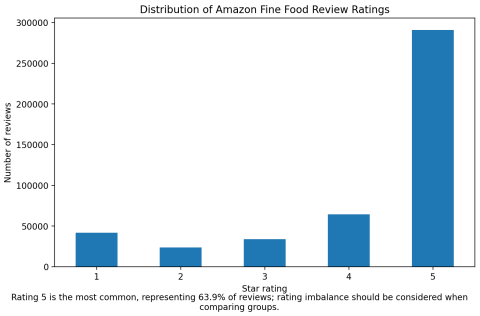


Figure 1: The number of five-star ratings is high: 290,934 out of 454,951 exploratory reviews (64%). It can be said that the rating is highly skewed.

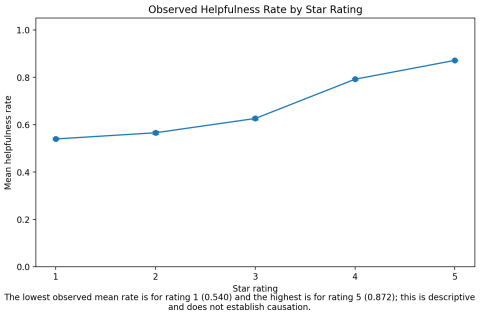


Figure 2: Helpfulness of mean reviews increases from 0.540 in one star review to 0.872 in five stars review. This is an unadjusted difference.

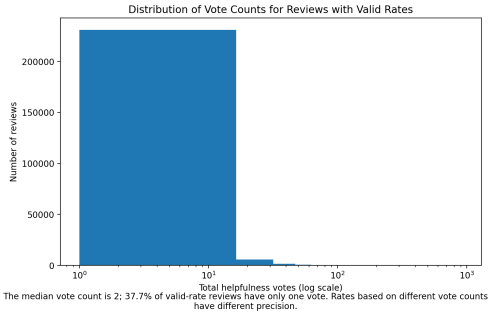


Figure 3: In terms of the total votes per review at the valid rate, the median is two, while 37.7% of observations have just one vote.

Basic questions

1. What is the distribution of review ratings, and how unbalanced are the rating categories?
2. How does helpfulness observed depend on the rating?
3. Does there exist a relation between helpfulness and length of reviews, and is this relation non-linear?
4. How does the variability of helpfulness depend on the number of votes?

Our problem statements

P1 (primary). In reviews that have valid helpfulness scores, is there any association between rating extremity and helpfulness, controlling for review length, summary length, and review year?

P2. Is there an association between review length and helpfulness, controlling for rating extremity, summary length, and review year, and is a nonlinear specification better suited to the data?

P3. Do text-based TF–IDF features improve predictive performance of held-out helpfulness ratings relative to review features?

What makes them novel: This study combines product-clustered explanation tests with a product-clustered hold-out test of structured and textual features.The innovation lies in this integrative, reproducible study of helpfulness correlation and prediction, not in the assertion that review helpfulness prediction or TF–IDF is a new research topic.

5. Statistical inferences

Tested Rows: The models used for explanation include 239,593 valid-rate reviews on 39,522 products, which include 802,496 helpful votes and 223,816 unhelpful votes. Test 4 includes 58,726 valid confirmation reviews on 9,794 products.

Test 1: Is rating extremity positively associated with helpfulness?

a) Hypotheses and planned method

H_0	$\beta_{\text{extremity}} = 0$
H_1	$\beta_{\text{extremity}} > 0$
Significance level	$\alpha = 0.05$; one-sided
Test	Grouped-binomial GLM with product-clustered standard errors
Practical benchmark	Odds ratio of 1.05 per one-unit increase in extremity

Rating extremity is $|\text{Score} - 3|$. The model adjusts for log review length, summary length and review year.

b) Other methods considered

Just a simple comparison between the average proportion of helpfulness would not take into consideration the number of votes corresponding to each proportion. Grouped-binomial method utilizes direct use of number of helpful and unhelpful votes. Product clustered method considers correlation of responses within each product; GEE method on product clusters was applied for sensitivity analysis despite its almost identical outcome.

c) Result

Estimate	$\beta = 0.3494$; odds ratio $= 1.418$ per unit
95% CI for odds ratio	1.382 to 1.455
Test statistic	$z = 26.70$
One-sided p-value	2.57×10^{-157}
Holm-adjusted p-value	5.13×10^{-157}
Decision	Reject H_0

The estimated relationship is positive and is larger than the odds-ratio threshold value at the point estimates. However, this does not imply causation.

Test 2: Is review length positively associated with helpfulness?

a) Hypotheses and planned method

H_0	$\beta_{\log(\text{length})} = 0$
H_1	$\beta_{\log(\text{length})} > 0$
Significance level	$\alpha = 0.05$; one-sided
Test	Grouped-binomial GLM with product-clustered standard errors
Practical benchmark	Odds ratio of 1.05 across the interquartile range

b) Other methods considered

The use of the raw length linear model would imply a constant rate of change with respect to additional characters. Log transformation would take care of the skew, while the spline model would be used to test for non-linearity. The joint product-clustered Wald test of the spline basis would determine whether it adds value together.

c) Result

Estimate	$\beta_{\log(\text{length})} = 0.4508$
95% CI for OR across length IQR	1.590 to 1.694
Test statistic	$z = 30.70$
One-sided p-value	2.70×10^{-207}
Holm-adjusted p-value	8.11×10^{-207}
Decision	Reject H_0

The adjusted association is positive and the point estimate exceeds the proposed practical benchmark. A clustered joint Wald test of four spline terms gave $\chi^2(4) = 1707.77$ with a p-value below numerical reporting precision. The ordinary likelihood-ratio comparison favored a spline by AIC, but its conventional p-value is not cluster-robust.

Test 3: Does summary length contribute explanatory information?

a) Hypotheses and method

H_0	$\beta_{\text{summary}} = 0$
H_1	$\beta_{\text{summary}} \neq 0$
Significance level	$\alpha = 0.05$; two-sided
Test	Product-clustered Wald test; nested-model likelihood-ratio comparison reported separately
Practical benchmark	1% relative deviance reduction

b) Other methods considered

The likelihood-ratio test compares the fits of two nested grouped binomial models; however, the traditional distribution used by the likelihood ratio test does not take into consideration the product clustering. The product-clustered Wald test is used in the Holm adjusted test; the likelihood-ratio test result is stated traditionally.

c) Result

Coefficient (per character)	-0.001671
95% CI for coefficient	$[-0.002673, -0.000669]$
Odds ratio per 10 characters	0.9834 (95% CI 0.9736–0.9933)
Wald statistic	$\chi^2(1) = 10.676$
Clustered p-value / Holm p-value	0.001085 / 0.001085
Conventional LR result	$\chi^2(1) = 96.424$, $p = 9.27 \times 10^{-23}$
Relative deviance reduction	0.0181%
Decision	Reject H_0 statistically; practical benchmark not reached

The negative adjusted correlation for summary length is relatively weak. The deviance reduction is very much below

the suggested 1% threshold.

5.1 Test 4: Does text improve prediction?

Question and hypotheses.

$$H_0 : \Delta\mathrm{MAE} = 0, \qquad H_1 : \Delta\mathrm{MAE} > 0,$$

where $\Delta\mathrm{MAE} = \mathrm{MAE}_{\mathrm{structured}} - \mathrm{MAE}_{\mathrm{text}}$. MAE reduction target via benchmarking is 0.01. For Ridge, baseline consists of rating, review length, summary length, and year, whereas comparison is made on the basis of additional 20,000 features, that is, TF-IDF unigrams and bigrams. Ridge value of $\alpha = 1.0$ is used for both cases. Predictions for both are truncated to $[0, 1]$. TF-IDF is trained using exploration data only.

Alternatives considered. RMSE is penalized for larger errors more heavily, while R^2 is less easily interpreted in terms of helpfulness-rate. The MAE is used primarily due to the ease of interpreting its absolute change. In case of a product, the product is resampled, and all reviews are kept.

Confirmation set	58,726 reviews; 9,794 products
Structured-only MAE	0.248705
Text-enhanced MAE	0.218671
Absolute / relative reduction	0.030034 / 12.08%
95% product-bootstrap CI	[0.02620, 0.03404]
Bootstrap replicates	2,000; seed 42
Practical benchmark	0.01; CI entirely above it

The text-augmented model yields better MAE for held-out products. The script outputs an empirical estimate of the bootstrap-tail probability at approximately 0.000500. As this is an estimation of the bootstrap tail probability and not a p-val- ue estimated under the null hypothesis, it should not be con- sidered a calibrated null hypothesis p-value. The confidence interval is the main source of uncertainty. This is predictive evidence, not causal evidence.

5.2 Multiplicity and inference audit

p-values after correction for Holm's method for Tests 1–3 is 5.13×10^{-157}, 8.11×10^{-207} and 0.001085respectively. The null hypothesis of all three tests is rejected at $\alpha = 0.05$ level. The Pearson dispersion ratio is approximately 2.12. While product-clustered analysis and GEE analyses account for some dependency problems, they do not completely preclude the problem of model misspecification. For Tests 1-2, the z statistics follow the asymptotic standard normal distribution, while the confidence intervals for the coefficients are calculated using the cluster-robust variance estimates. The Clustered Wald statistic for Test 3 has an asymptotic distribution of $\chi^2(1)$.

6. Limitations

- **No causality; correlation.** Reviews' usefulness ratings and lengths, as well as helpfulness votes are observational data, and there may be other variables which affect both.
- **Uneven vote counts.** There are many records that contain no votes at all or very low numbers of votes. Reviews with zero votes have no defined usefulness scores.
- **Dependence and overdispersion.** Reviews could be characterized by common products, customers, or any other factors. Standard errors with product clustering help to take into account dependence within the product, but a Pearson ratio of around 2.12 reveals overdispersion.
- **Quality of data.** Two invalid useful responses were marked and removed from the valid rate analysis. Long reviews were kept since an IQR outlier is not considered an error on its own.
- **Generalization.** The set of predictions confirmed in the confirmation phase includes products which are not included in the exploration process, but still belong to the same source data set. The results are not generalizable across other platforms or domains of products.
- **Model selection and threshold settings.** The comparative prediction experiment utilizes a constant Ridge regularization term and TF-IDF setting. Any practical threshold values cannot be considered preregistered unless there is proof of their time point.

Conclusion

Rating extremity and review length exhibit a statistically positive relation to helpfulness. The length of summary is statistically significant. It does not help reduce deviance much. The addition of TF–IDF based predictors reduces the MAE for the confirmation set by 12.1%. These results show the power of predictions using reviews. Illustrate the difference between statistical significance, practical significance and prediction, versus causation.

Data and code

Dataset: Stanford SNAP Amazon Fine Food Reviews, `https://snap.stanford.edu/data/web-FineFoods.html`. Project README includes the documentation for the scripts, output, and data preparation process. Substitute the placeholder for the repository URL at the beginning of this report with the publicly accessible GitHub URL once the project is uploaded.